

FINITE BLOCK EXHAUSTION AND ROOTED OCCUPANCY IN THE INVERSE COLLATZ SYSTEM A NO-ALTERNATIVE PROOF VIA TERNARY COUNTERS, PRIMITIVE CARRY, AND FORCED THREAD DISPLACEMENT

MICHAEL E. SPENCER

ABSTRACT. The odd Collatz inverse system is studied through finite refinement blocks. The central object is not an asymptotic density estimate, but an exact occupancy counter. At ternary depth J , the rooted admissible counter has 3^J cells, 2^J carried cells, and $3^J - 2^J$ occupied cells. In the corresponding six-case ordinal counter, the split is $6 = 3 + 2 + 1$: three integer-producing cases, two carried cases, and one excluded case. These finite identities are supported by a machine-checked formalization of the combinatorial core.

The paper then ties the counter to the inverse Collatz arithmetic. The local affine dyadic atlas partitions the odd integers by the unique valuation $k = \nu_2(3m + 1)$. Primitive and carried representatives are separated by the identity $3n/3^{J+1} = n/3^J$, so carried cells are lower-level copies rather than new gaps. Fixed inverse words act affinely on residue threads. Neighboring thread positions are displaced after one traversal by a pure dyadic gap 2^{S+1} , which is a unit modulo every power of 3. A word that must restart a nontrivial cycle must return to the same live residue class modulo 6, but forced dyadic displacement prevents stationarity in the triadic refinement tower.

The main theorem is an exhaustion of alternatives. A missing component outside rooted occupancy must be primitive, carried, cyclic, or overlapping. Primitive gaps contradict exact finite-block occupancy together with disjoint primitive placement. Carried gaps are lower-scale representatives. Cyclic alternatives are excluded by forced thread displacement. Overlaps are excluded by unique forward parentage. Hence no admissible alternative remains outside the rooted occupancy. The final section translates rooted inverse occupancy into forward odd convergence and then to all positive integers, since every even integer reaches an odd integer by finitely many divisions by 2.

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1. INTRODUCTION

The Collatz problem asks whether every positive integer reaches 1 under the usual iteration: an even integer is divided by 2, and an odd integer m is sent to $3m + 1$, after which the even factors are removed. The odd-to-odd map is

$$T(m) = \frac{3m + 1}{2^{\nu_2(3m+1)}}.$$

The corresponding inverse branch is

$$R(n; k) = \frac{2^k n - 1}{3},$$

where admissibility is the congruence

$$2^k n \equiv 1 \pmod{3}.$$

The proof developed here treats the inverse system as a finite-block occupancy problem. The refinement moduli are

$$M_J = 2 \cdot 3^J.$$

A fixed inverse word occupies one residue thread modulo M_J ; the full occupancy structure is obtained by taking the finite collection of all compatible word-cells at level J and separating primitive representatives from carried representatives. The main result is an exhaustion theorem: once exact block occupancy, carry consistency, disjoint placement, and forced thread displacement are in place, no alternative component remains.

This is not a proof by limiting density. The limiting identities appear only after exact finite equalities are established. The core finite count is

$$|\mathcal{W}_J| = 3^J, \quad |\mathcal{C}_J| = 2^J, \quad |\mathcal{O}_J| = 3^J - 2^J.$$

The carried cells are not missing cells. They are the unresolved representatives that pass to the next refinement level. The primitive cells are the new cells at that scale. A least missing representative would therefore have to be primitive or carried. The primitive case contradicts exact finite occupancy and disjoint placement; the carried case descends to a lower representative. Cyclic and overlapping alternatives are excluded by the fixed-word displacement law.

2. PRIOR WORK AND POSITION OF THE PRESENT ARGUMENT

The $3n + 1$ problem has been studied through residue classes, inverse predecessor trees, parity vectors, dyadic valuations, probabilistic models, and dynamical graph structure. The present paper uses several standard arithmetic ingredients from this literature, but arranges them into a finite-block exhaustion argument. This section records the relation between the present method and prior work, and isolates the specific point at which the argument here differs from earlier approaches.

The inverse odd Collatz step is

$$R(n; k) = \frac{2^k n - 1}{3}.$$

Its basic admissibility condition is

$$2^k n \equiv 1 \pmod{3}.$$

Thus the parity of k is determined by the residue of n modulo 3. The live primitive odd classes are represented by 1 (mod 6) and 5 (mod 6), while the class 3 (mod 6) is carried rather than primitive. Congruence restrictions of this kind, including the use of residue classes modulo 6, 18, and higher

powers of 3, appear throughout earlier inverse and cycle analyses of the problem; see Everett [1], Garner [2], and the surveys of Lagarias [3, 4]. In the present paper, these congruence restrictions are not used only as local filters. They are assembled into the finite refinement tower

$$M_J = 2 \cdot 3^J,$$

where each finite level separates primitive representatives from carried representatives.

Dyadic valuation is also a standard organizing device. For an odd integer m , the value

$$k = \nu_2(3m + 1)$$

places m into a unique dyadic slice. The slice with fixed k has relative density 2^{-k} among odd integers, a fact used frequently in heuristic and probabilistic treatments of the problem; see Lagarias [3, 4] and Tao [5]. In the present paper, this valuation is used arithmetically rather than probabilistically. Each slice is represented by the affine formula

$$m = 2^{k+1}t + \frac{2^k x_k - 1}{3},$$

where

$$x_k = \begin{cases} 5, & k \text{ odd}, \\ 1, & k \text{ even}. \end{cases}$$

Thus the valuation parameter k supplies a disjoint affine atlas of the odd integers, not merely a density weight.

The inverse graph viewpoint has also been developed systematically. Wirsching treats the dynamical system generated by the $3n + 1$ map through predecessor sets, inverse trees, density functions, and graph-theoretic structure [6]. The present work remains within the inverse-graph perspective, but replaces asymptotic predecessor-density reasoning with a finite-block occupancy argument. At refinement depth J , the rooted ternary counter has exact finite cardinalities

$$|\mathcal{W}_J| = 3^J, \quad |\mathcal{C}_J| = 2^J, \quad |\mathcal{O}_J| = 3^J - 2^J.$$

Here $\mathcal{W}_J$ is the full rooted ternary word block, $\mathcal{C}_J$ is the carried part, and $\mathcal{O}_J$ is the occupied part. Hence the finite rooted occupancy ratio is

$$\frac{|\mathcal{O}_J|}{|\mathcal{W}_J|} = 1 - \left(\frac{2}{3}\right)^J.$$

The argument is therefore stated in exact finite-block form rather than as a limiting density assertion.

Affine lifts of the form

$$m \mapsto 4m + 1$$

arise naturally in inverse Collatz constructions. If an admissible exponent is raised by two, then

$$\frac{2^{k+2}n - 1}{3} = 4 \left(\frac{2^k n - 1}{3} \right) + 1.$$

Earlier uses of this relation describe backward progressions, special families, or parity-vector behavior. Here the same recurrence is used as part of a finite-block occupancy structure. The affine rails are disjoint, their positions agree with dyadic valuation, and their carried representatives are tracked through the refinement tower.

The main distinction of the present paper is the use of an exhaustion-of-alternatives proof. The proof combines four mechanisms. First, the finite rooted counter gives the exact occupancy

$$|\mathcal{O}_J| = 3^J - 2^J.$$

Second, the primitive/carry decomposition identifies carried representatives as lower-level copies:

$$\frac{3n}{3^{J+1}} = \frac{n}{3^J}.$$

Thus a carried representative is not a new primitive cell. Third, fixed admissible words act on residue threads by affine maps. If K has length j and total dyadic weight $S(K)$, then

$$F_K(n) = \frac{2^{S(K)}n - B_K}{3^j}.$$

For neighboring positions on the same fixed-word thread,

$$n_t = r(K) + 2 \cdot 3^j t,$$

one obtains

$$F_K(n_{t+h}) - F_K(n_t) = h \cdot 2^{S(K)+1}.$$

Thus the triadic thread spacing is consumed by the word and converted into pure dyadic displacement. Fourth, this displacement is a unit modulo every power of 3:

$$\gcd(2^{S(K)+1}, 3^q) = 1 \quad (q \geq 0).$$

Therefore a nontrivial fixed word cannot restart indefinitely on the same ternary refinement phase.

These mechanisms exclude every admissible alternative to rooted occupancy. A supposed missing representative would have to be primitive, carried, cyclic, or overlapping. A primitive gap contradicts exact finite-block occupancy together with disjoint primitive placement. A carried gap is already a lower-level representative and therefore cannot be minimal. A cyclic alternative is excluded by forced dyadic displacement through the ternary refinement tower. An overlap is excluded by unique valuation and disjoint placement. Consequently, no admissible alternative component remains outside the rooted inverse occupancy structure.

Once rooted odd occupancy has been established, the passage to the usual forward Collatz map is standard. If

$$m = \frac{2^k n - 1}{3},$$

then

$$3m + 1 = 2^k n,$$

so the odd-to-odd forward map sends m back to n . Hence a finite rooted inverse address gives a finite forward descent to 1. Since every positive even integer reaches an odd integer after finitely many divisions by 2, the odd result extends to all positive integers.

3. MAIN THEOREM AND DEPENDENCY MAP

Theorem 3.1 (Finite block exhaustion of alternatives). *The admissible inverse Collatz structure rooted at 1 exhausts the odd integers. No admissible primitive gap, carried gap, cyclic component, or overlap component remains outside the rooted occupancy.*

The proof is assembled from the following dependencies.

- (i) **Local affine dyadic atlas.** Every odd integer belongs to a unique affine slice determined by $k = \nu_2(3m + 1)$.
- (ii) **Finite rooted counter.** At depth J , the rooted ternary counter has $3^J - 2^J$ occupied cells and 2^J carried cells.
- (iii) **Primitive/carry separation.** A carried representative $3n/3^{J+1}$ is identical to the lower representative $n/3^J$ and is not a new primitive cell.
- (iv) **Fixed-word displacement.** A word of total dyadic weight S sends neighboring positions on its thread to endpoints separated by 2^{S+1} .
- (v) **Ternary-unit obstruction.** The displacement 2^{S+1} is coprime to every 3^q , so it cannot vanish in the triadic refinement tower.
- (vi) **Modulo-six restart obstruction.** A repeated word must end in the same live residue class modulo 6 required by its first exponent; otherwise its first exponent is no longer admissible.

- (vii) **No-minimal-gap criterion.** Finite scale matching and injective primitive placement force all primitive target cells to be occupied.

The conclusion is direct. A supposed outside component has one of the following forms: primitive gap, carried gap, cyclic component, or overlap. These alternatives are exhausted and excluded.

4. THE ODD INVERSE SYSTEM

Definition 4.1 (Odd map and inverse branch). For odd m , define

$$T(m) = \frac{3m+1}{2^{\nu_2(3m+1)}}.$$

For an odd n and an integer $k \geq 1$, define the inverse branch

$$R(n; k) = \frac{2^k n - 1}{3},$$

when the right-hand side is an integer.

Definition 4.2 (Live classes). The live same-scale inverse classes are

$$1 \pmod{6}, \quad 5 \pmod{6}.$$

The class $3 \pmod{6}$ is divisible by 3 and is carried to a different triadic scale.

Lemma 4.3 (Admissible parity). *Let n be odd.*

- (a) *If $n \equiv 1 \pmod{6}$, then admissible exponents are even.*
- (b) *If $n \equiv 5 \pmod{6}$, then admissible exponents are odd.*
- (c) *If $n \equiv 3 \pmod{6}$, then no same-scale inverse exponent is admissible.*

Proof. Modulo 3, $2^k \equiv 1$ for even k and $2^k \equiv 2$ for odd k . The condition $2^k n \equiv 1 \pmod{3}$ therefore fixes the parity of k according to $n \pmod{3}$. If $3 \mid n$, then $2^k n \equiv 0 \pmod{3}$ and the congruence cannot hold. $\square$

5. THE LOCAL AFFINE DYADIC ATLAS

The following direct atlas is the local exhaustion of the odd integers. It is the arithmetic counterpart of the dyadic valuation table.

For $k \geq 1$, set

$$x_k = \begin{cases} 5, & k \text{ odd,} \\ 1, & k \text{ even,} \end{cases} \quad a_k = \frac{2^k x_k - 1}{3}.$$

Define

$$\mathcal{S}_k(t) = 2^{k+1}t + a_k, \quad t \geq 0.$$

Thus

$$\mathcal{S}_1(t) = 4t + 3, \quad \mathcal{S}_2(t) = 8t + 1, \quad \mathcal{S}_3(t) = 16t + 13, \quad \mathcal{S}_4(t) = 32t + 5.$$

Proposition 5.1 (Affine dyadic partition). *The odd integers are partitioned by the slices*

$$\mathbb{N}_{\text{odd}} = \bigsqcup_{k \geq 1} \mathcal{S}_k.$$

The k -slice has density 2^{-k} among odd integers, and

$$\sum_{k=1}^{\infty} 2^{-k} = 1.$$

Proof. Given odd m , the integer

$$k = \nu_2(3m + 1)$$

is unique. Write

$$3m + 1 = 2^k n$$

with n odd. Then

$$m = \frac{2^k n - 1}{3}.$$

The admissible parity lemma forces $n \equiv x_k \pmod{6}$, so $n = 6t + x_k$ for a unique $t \geq 0$. Substitution gives

$$m = \frac{2^k(6t + x_k) - 1}{3} = 2^{k+1}t + \frac{2^k x_k - 1}{3}.$$

Thus every odd integer belongs to exactly one slice. Since $\mathcal{S}_k$ is one residue class modulo 2^{k+1} , its density among all positive integers is $2^{-(k+1)}$, and its relative density among odd integers is 2^{-k} . $\square$

Lemma 5.2 (Forward coalescence). *If*

$$m = \frac{2^k n - 1}{3}$$

with m, n odd, then

$$T(m) = n.$$

Proof. The equation gives

$$3m + 1 = 2^k n.$$

Since n is odd, $\nu_2(3m + 1) = k$, hence

$$T(m) = \frac{3m + 1}{2^k} = n.$$

$\square$

Corollary 5.3 (Unique immediate parent). *Every odd integer has a unique immediate parent under T .*

Proof. The valuation $\nu_2(3m + 1)$ is unique, and then $T(m)$ is forced. $\square$

6. ROOTED TERNARY OCCUPANCY COUNTER

Definition 6.1 (Ternary alphabet). Let

$$\mathcal{A}_3 = \{\mathsf{T}, \mathsf{C}_0, \mathsf{C}_1\}.$$

Here T denotes terminal or occupied, while C_0 and C_1 are carried symbols.

Definition 6.2 (Words, carried cells, occupied cells). For $J \geq 0$, set

$$\mathcal{W}_J = \mathcal{A}_3^J.$$

The carried cells are

$$\mathcal{C}_J = \{w \in \mathcal{W}_J : w_i \in \{\mathsf{C}_0, \mathsf{C}_1\} \text{ for every } i\},$$

and the occupied cells are

$$\mathcal{O}_J = \mathcal{W}_J \setminus \mathcal{C}_J.$$

Theorem 6.3 (Finite block rooted occupancy). *For every $J \geq 0$,*

$$|\mathcal{W}_J| = 3^J, \quad |\mathcal{C}_J| = 2^J, \quad |\mathcal{O}_J| = 3^J - 2^J.$$

Therefore

$$\frac{|\mathcal{O}_J|}{|\mathcal{W}_J|} = 1 - \left(\frac{2}{3}\right)^J.$$

Proof. There are three choices at each position of a word in $\mathcal{W}_J$, so $|\mathcal{W}_J| = 3^J$. A carried word has two choices at each position, so $|\mathcal{C}_J| = 2^J$. Since $\mathcal{O}_J$ is the complement of $\mathcal{C}_J$ in $\mathcal{W}_J$,

$$|\mathcal{O}_J| = 3^J - 2^J.$$

The ratio follows by division by 3^J . □

Definition 6.4 (First occupied level). For $w \in \mathcal{O}_J$, let $\text{first}(w)$ be the least position j such that $w_j = \mathsf{T}$.

Lemma 6.5 (Exact-level contribution). For $1 \leq j \leq J$, the number of words $w \in \mathcal{W}_J$ with $\text{first}(w) = j$ is

$$2^{j-1}3^{J-j}.$$

Consequently,

$$\sum_{j=1}^J 2^{j-1}3^{J-j} = 3^J - 2^J.$$

Proof. The first $j-1$ positions must be carried, giving 2^{j-1} choices. The j th position is fixed to be T . The remaining $J-j$ positions are arbitrary ternary symbols. These classes partition $\mathcal{O}_J$ by first occupied level, so the sum is $|\mathcal{O}_J| = 3^J - 2^J$. □

7. SIX-CASE ORDINAL COUNTER

Definition 7.1 (Six-case alphabet). Let

$$\mathcal{A}_6 = \{\mathsf{T}_1, \mathsf{T}_2, \mathsf{T}_3, \mathsf{C}_0, \mathsf{C}_1, \mathsf{E}\}.$$

The symbols $\mathsf{T}_1, \mathsf{T}_2, \mathsf{T}_3$ are integer-producing cases; $\mathsf{C}_0, \mathsf{C}_1$ are carried cases; E is excluded.

Definition 7.2 (Six-case decomposition). Let $\mathcal{W}_J^{(6)} = \mathcal{A}_6^J$. Let $\mathcal{C}_J^{(6)}$ be the words containing only carried symbols. Let $\mathcal{O}_J^{(6)}$ be the words whose first non-carried symbol is one of $\mathsf{T}_1, \mathsf{T}_2, \mathsf{T}_3$. Let $\mathcal{X}_J^{(6)}$ be the words whose first non-carried symbol is E .

Theorem 7.3 (Six-case finite decomposition). For every $J \geq 1$,

$$\begin{aligned} |\mathcal{C}_J^{(6)}| &= 2^J, \\ |\mathcal{O}_J^{(6)}| &= \frac{3}{4}(6^J - 2^J), \\ |\mathcal{X}_J^{(6)}| &= \frac{1}{4}(6^J - 2^J). \end{aligned}$$

Equivalently,

$$\begin{aligned} \frac{|\mathcal{O}_J^{(6)}|}{6^J} &= \frac{3}{4} \left(1 - \frac{1}{3^J} \right), \\ \frac{|\mathcal{X}_J^{(6)}|}{6^J} &= \frac{1}{4} \left(1 - \frac{1}{3^J} \right), \quad \frac{|\mathcal{C}_J^{(6)}|}{6^J} = \frac{1}{3^J}. \end{aligned}$$

Proof. The all-carried count is 2^J . If the first non-carried symbol occurs at level j , then the first $j-1$ positions give 2^{j-1} carried choices. For occupied words there are 3 choices for the first non-carried symbol, and the remaining $J-j$ positions are arbitrary six-case symbols. Hence

$$|\mathcal{O}_J^{(6)}| = \sum_{j=1}^J 3 \cdot 2^{j-1} 6^{J-j} = \frac{3}{4}(6^J - 2^J).$$

For excluded words, replace the coefficient 3 by 1:

$$|\mathcal{X}_J^{(6)}| = \sum_{j=1}^J 2^{j-1} 6^{J-j} = \frac{1}{4}(6^J - 2^J).$$

The ratios follow by division by 6^J . □

8. PRIMITIVE AND CARRIED REPRESENTATIVES

Definition 8.1 (Odd residue block). At scale J , let

$$B_J = \{r \bmod 2 \cdot 3^J : r \text{ is odd}\}.$$

Then $|B_J| = 3^J$.

Definition 8.2 (Primitive and carried slots). For $J \geq 1$, define

$$\mathcal{P}_J = \{r \in B_J : 3 \nmid r\}, \quad D_J = \{r \in B_J : 3 \mid r\}.$$

The elements of $\mathcal{P}_J$ are primitive at scale J , and the elements of D_J are carried.

Lemma 8.3 (Primitive-carried split). For $J \geq 1$,

$$|D_J| = 3^{J-1}, \quad |\mathcal{P}_J| = 2 \cdot 3^{J-1}.$$

Proof. Each carried residue is uniquely of the form $3r$, where r is an odd residue modulo $2 \cdot 3^{J-1}$. Therefore $|D_J| = 3^{J-1}$. Since $|B_J| = 3^J$,

$$|\mathcal{P}_J| = 3^J - 3^{J-1} = 2 \cdot 3^{J-1}.$$

□

Lemma 8.4 (Carry identification). *Division by 3 gives a bijection*

$$D_J \longrightarrow B_{J-1}.$$

Equivalently,

$$\frac{3n}{3^J} = \frac{n}{3^{J-1}}.$$

Proof. Multiplication by 3 sends an odd residue modulo $2 \cdot 3^{J-1}$ to a carried odd residue modulo $2 \cdot 3^J$, and division by 3 reverses this operation. □

Corollary 8.5 (Carried cells are lower representatives). *A carried cell at scale J is not a new gap. It is the same representative already present at scale $J - 1$.*

Proof. This is exactly the carry identification. □

9. FIXED-WORD THREADS AND DYADIC DISPLACEMENT

Definition 9.1 (Directed word). A directed word is a finite sequence

$$K = (k_0, k_1, \dots, k_{j-1}).$$

Its length is j , and its total dyadic weight is

$$S(K) = k_0 + k_1 + \dots + k_{j-1}.$$

Definition 9.2 (Affine word action). The inverse composition determined by K has affine form

$$F_K(n) = \frac{2^{S(K)}n - B_K}{3^j},$$

where B_K is the correction term determined by the word.

Definition 9.3 (Fixed-word thread). Let $M_j = 2 \cdot 3^j$. A fixed-word thread for K is an affine residue class

$$\Lambda_K = \{n_t = r(K) + M_j t : t \geq 0\}$$

on which the word K is admissible.

Lemma 9.4 (Pure dyadic image separation). *Let $n_t = r(K) + 2 \cdot 3^j t$. For every $h \geq 1$,*

$$F_K(n_{t+h}) - F_K(n_t) = h \cdot 2^{S(K)+1}.$$

In particular,

$$F_K(n_{t+1}) - F_K(n_t) = 2^{S(K)+1}.$$

Proof. Since

$$n_{t+h} - n_t = 2 \cdot 3^j h,$$

the affine form gives

$$F_K(n_{t+h}) - F_K(n_t) = \frac{2^{S(K)}}{3^j} (2 \cdot 3^j h) = h \cdot 2^{S(K)+1}.$$

□

Lemma 9.5 (Dyadic displacement is a ternary unit). *For every $S \geq 0$ and every $q \geq 1$,*

$$\gcd(2^{S+1}, 3^q) = 1.$$

Proof. The first integer is a power of 2, and the second is a power of 3. □

10. FORCED THREAD DISPLACEMENT AND ACYCLICITY

Definition 10.1 (Cycle-compatible restart). A word K is cycle-compatible on a thread when the endpoint of one traversal lies in the same live residue class modulo 6 required to apply the first exponent of K again.

Lemma 10.2 (Modulo-six restart obstruction). *A non-multiple-of-3 endpoint can restart the same word only when it returns to the same live residue class modulo 6 as its starting position. The opposite live class reverses the admissible parity of the first exponent, while $3 \pmod{6}$ is not a same-scale live class.*

Proof. The admissible parity lemma determines the required parity of the first exponent from the live class modulo 6. Switching between $1 \pmod{6}$ and $5 \pmod{6}$ reverses the required parity. Landing in $3 \pmod{6}$ leaves the same-scale live inverse system. □

Theorem 10.3 (Forced thread displacement excludes nontrivial cycle words). *No nonempty fixed word can coherently restart as a nontrivial cycle word on its own thread while preserving the required live residue class modulo 6.*

Proof. A nontrivial cycle word must restart from the same live residue class modulo 6; otherwise the first exponent of the repeated word is not admissible. Thus the endpoint copy must occupy the same compatible triadic refinement phase required to begin the word again.

Neighboring realizations of the same word are separated after one traversal by the pure dyadic gap $2^{S(K)+1}$. This gap is a unit modulo 3^q for every $q \geq 1$, so it never vanishes in the triadic refinement tower. A displaced copy therefore cannot remain stationary in the required triadic phase at all levels. The same word cannot both return to its own restart phase and be displaced by a nonzero ternary unit. Hence no nontrivial cycle word exists. □

Corollary 10.4 (No cyclic alternative). *There is no nontrivial cyclic component compatible with the fixed-word residue law.*

Proof. A cyclic component would contain a finite returning word. Such a word is excluded by forced thread displacement. □

11. DISJOINT PRIMITIVE PLACEMENT

Proposition 11.1 (Disjoint primitive placement). *Distinct primitive occupied cells cannot represent the same dependency position.*

Proof. If two primitive occupied cells represented the same odd value m , then unique immediate parentage forces the same $k = \nu_2(3m + 1)$ and the same parent $T(m)$. Hence they coincide at the local affine slice level.

If the two descriptions were separated by longer finite words before coalescing, their coalescence would produce a finite returning word between copies of the same residue structure. That would be a cyclic alternative, excluded by forced thread displacement. Thus primitive placement is disjoint. $\square$

12. SCALE MATCHING AND FINITE OCCUPANCY

Definition 12.1 (Primitive target block). At scale J , the primitive target block $\mathcal{P}_J$ is the set of primitive odd representatives after carried duplicates have been removed.

Proposition 12.2 (Scale matching). *For every finite scale J ,*

$$|\mathcal{O}_J| = |\mathcal{P}_J| = 3^J - 2^J.$$

Proof. The rooted occupied count is $|\mathcal{O}_J| = 3^J - 2^J$. The primitive target block is obtained from the same finite refinement block after removing the carried addresses, since carried addresses are lower-level representatives. Therefore the target capacity is the total ternary block count 3^J minus the carried count 2^J , giving

$$|\mathcal{P}_J| = 3^J - 2^J.$$

$\square$

Corollary 12.3 (Finite scale primitive surjectivity). *At every finite scale J , rooted placement occupies all primitive target cells of $\mathcal{P}_J$.*

Proof. Disjoint primitive placement gives injectivity of the rooted placement into the primitive target block. Scale matching gives equal finite cardinality. An injective map between finite sets of equal cardinality is surjective. $\square$

13. NO MINIMAL PRIMITIVE GAP

Definition 13.1 (Primitive gap). A primitive gap is a primitive odd representative not occupied by the rooted address structure at its finite scale.

Theorem 13.2 (No minimal primitive gap). *There is no least unoccupied primitive odd representative.*

Proof. Assume a primitive gap exists and choose the least one by finite scale, then by order inside that scale. If the representative is carried, then carry identification rewrites it as a lower-level representative, contradicting minimality. If the representative is primitive, then it belongs to $\mathcal{P}_J$ for some finite scale J . Finite scale primitive surjectivity says that all cells of $\mathcal{P}_J$ are occupied. This contradicts the choice of a primitive gap. $\square$

Corollary 13.3 (No primitive gaps). *The primitive gap set is empty.*

Proof. If it were nonempty, the well-ordering of finite-scale representatives would provide a least primitive gap. This contradicts the previous theorem. $\square$

14. EXHAUSTION OF ALTERNATIVES

Theorem 14.1 (No-alternative exhaustion). *No admissible component remains outside rooted occupancy. There is no primitive gap component, no carried gap component, no overlap component, and no cyclic component.*

Proof. Every alternative outside rooted occupancy must fall into one of four cases.

First, it may be primitive. This is impossible because there are no primitive gaps.

Second, it may be carried. But a carried representative is identified with a lower-level representative by

$$\frac{3n}{3^{J+1}} = \frac{n}{3^J}.$$

Thus it is not a new outside cell.

Third, it may overlap a rooted cell while claiming a different dependency. This contradicts unique immediate parentage and disjoint primitive placement.

Fourth, it may form a cyclic alternative. This is excluded by forced thread displacement: a finite returning word would have to preserve the same live residue class modulo 6 while remaining stationary in the triadic refinement tower, but its copy displacement is a dyadic unit modulo every power of 3.

All alternatives are exhausted and excluded. $\square$

Theorem 14.2 (Rooted occupancy). *The rooted inverse occupancy exhausts the primitive odd representatives, with carried representatives identified by their lower-level copies. Hence the rooted dependency structure has no admissible outside component.*

Proof. Finite scale primitive surjectivity occupies all primitive target cells at every scale. Carry identification removes duplicate carried representatives. Disjoint primitive placement prevents overlap. Forced thread displacement excludes cyclic alternatives. By no-alternative exhaustion, no admissible outside component remains. $\square$

15. FORWARD-INVERSE EQUIVALENCE ON THE ODD SYSTEM

Definition 15.1 (Rooted inverse address). An odd integer m has a rooted inverse address if there is a finite admissible word $K = (k_0, \dots, k_{j-1})$ such that

$$m = F_K(1).$$

Lemma 15.2 (Forward deletion of an inverse address). *If $m = F_K(n)$ for a finite admissible word K , then applying the corresponding odd forward map deletes the word and returns to n after finitely many odd steps.*

Proof. For a one-step word this is exactly forward coalescence. For a longer word, write $K = K'k$ and set $u = F_{K'}(n)$. Then

$$F_K(n) = \frac{2^{k_u} - 1}{3}.$$

By forward coalescence, this maps forward to u . Repeating the argument deletes one inverse step at a time until the value returns to n . $\square$

Theorem 15.3 (Odd convergence). *Every positive odd integer reaches 1 under the odd-to-odd Collatz map T .*

Proof. By rooted occupancy, every odd integer has a finite rooted inverse address from 1. Forward deletion of the inverse address removes one symbol at a time and returns the value to 1 after finitely many odd steps. $\square$

16. REDUCTION OF EVEN INTEGERS

Lemma 16.1 (Even reduction to an odd integer). *Every positive integer N can be written uniquely as*

$$N = 2^a m,$$

where $a \geq 0$ and m is odd. Repeated division by 2 sends N to m in a steps.

Proof. This is the standard factorization of a positive integer by its 2-adic valuation. $\square$

Theorem 16.2 (Full Collatz convergence). *Every positive integer reaches 1 under the Collatz iteration.*

Proof. Let $N > 0$. Write $N = 2^a m$ with m odd. Repeated even steps send N to m . By odd convergence, m reaches 1 under finitely many odd-to-odd Collatz steps. Therefore N reaches 1. $\square$

17. MACHINE-CHECKED SUPPORT LEMMAS

The finite combinatorial and structural backbone of this paper has a corresponding Lean formalization. The verified declarations include the following.

- `card_word3`: $|\mathcal{W}_J| = 3^J$.
- `card_carried3`: $|\mathcal{C}_J| = 2^J$.
- `card_occupied3`: $|\mathcal{O}_J| = 3^J - 2^J$.
- `ratio_occupied3`: $|\mathcal{O}_J|/|\mathcal{W}_J| = 1 - (2/3)^J$.
- `card_firstAt3` and `sum_firstLevel3`: exact-level ternary occupancy and the geometric identity.
- `card_carried6`, `card_occupied6`, `card_excluded6`, and `decomposition6`: the six-case ordinal counter.
- `card_oddResBlock`, `card_carrySlots`, `card_primSlots`, `carry_image`, and `carry_injOn`: primitive/carry counts and carry injection.
- `dyadic_separation` and `dyadic_separation_neighbour`: fixed-word dyadic thread displacement.
- `restart_parity_one` and `restart_parity_five`: the modulo-six restart obstruction.
- `displacement_unit_mod_three`: 2^{S+1} is coprime to every 3^q .
- `no_minimal_gap` and `no_minimal_gap_no_min`: the finite-cardinality no-minimal-gap criterion from injectivity and equal cardinality.

These results verify the finite occupancy counter, six-case counter, primitive/carry representative structure, fixed-thread displacement, ternary-unit obstruction, and finite no-minimal-gap backbone used in the exhaustion proof.

18. CONCLUSION

The proof is an exhaustion of alternatives. The finite ternary counter gives exact rooted occupancy,

$$|\mathcal{O}_J| = 3^J - 2^J.$$

The carry law identifies carried representatives as lower-level copies,

$$\frac{3n}{3^{J+1}} = \frac{n}{3^J}.$$

Fixed-word threads transform neighboring triadic spacing into pure dyadic displacement,

$$F_K(n_{t+1}) - F_K(n_t) = 2^{S(K)+1},$$

and this displacement is a unit modulo every power of 3.

Thus a missing alternative cannot be carried, because carried representatives descend. It cannot be primitive, because finite scale matching and disjoint primitive placement fill the primitive target

block. It cannot be cyclic, because forced thread displacement prevents coherent same-word restart. It cannot be overlap, because unique parentage and forced displacement exclude incompatible coalescence. Therefore no admissible alternative component remains outside rooted occupancy.

Rooted occupancy gives finite inverse addresses for all odd integers. Forward iteration deletes those addresses one step at a time and returns every odd integer to 1. Every positive even integer reaches an odd integer after finitely many divisions by 2. Hence every positive integer reaches 1.

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